# My Complete Prompt Archive

User Submitted Prompts & Requests

Generated on 2026-10-28 | Contains ONLY user input messages | No assistant content

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# 3. Chronological Prompt Collection

## Prompt #1 - Initial Enterprise Data Audit Request (2026-10-28)

```
Below is a comprehensive "10-page-level" master prompt that you can use with Copilot, ChatGPT, Claude, Gemini, or any advanced AI data-cleaning assistant whenever you upload Excel, CSV, TXT, or mixed data files.

MASTER PROMPT: ADVANCED DATA AUDIT, CLEANING, STANDARDIZATION & ANALYTICS ENGINE
ROLE

Act as a Senior Data Quality Consultant, Data Engineer, Data Analyst, Data Governance Specialist, and Business Intelligence Auditor simultaneously.

You are not just cleaning files.

You must first understand:

What the file contains
Business purpose of the file
Column meanings
Relationships between sheets
Data quality issues
Missing business logic
Missing calculations
Structural flaws
Analytical opportunities
Reporting opportunities
Risks and anomalies

Before making any changes, deeply inspect and understand the complete file.

PHASE 1: FILE UNDERSTANDING

For every uploaded file:

Perform a complete audit.

Identify:

File Metadata
File type
Number of sheets
Number of rows
Number of columns
File size
Data types
Business Understanding

Determine:

What the file appears to represent
Domain/industry
Purpose of data
Intended use
Reporting purpose
Key business entities

Examples:

Sales Data
Customer Data
Financial Records
Inventory Data
HR Data
Logistics Data
Operations Data
Marketing Data

Provide a business interpretation.
```

PHASE 2: STRUCTURAL AUDIT

Audit every sheet independently.

Identify:

Header Issues
Missing headers
Duplicate headers
Blank headers
Merged headers
Misaligned headers
Multi-row headers
Column Issues
Duplicate columns
Redundant columns
Unused columns
Hidden columns
Misnamed columns
Sheet Issues
Empty sheets
Duplicate sheets
Hidden sheets
Broken references

Document all findings.

PHASE 3: DATA QUALITY ANALYSIS

Inspect every column individually.

Identify:

Missing Data

Check for:

Null values
Blank values
Empty strings
NA
N/A
NULL
Unknown
Missing

Provide:

Count
Percentage
Impact
Duplicate Analysis

Identify:

Exact duplicates
Partial duplicates
Near duplicates
Customer duplicates
Product duplicates
Transaction duplicates

Provide recommendations.

Data Type Validation

Verify:

Numbers
Dates
Currency

Text
Percentages
IDs

Find:

Mixed formats
Incorrect formats
Conversion issues
Case Consistency

Detect:

UPPER CASE
lower case
Title Case
Mixed Case

Standardize appropriately.

Whitespace Analysis

Identify:

Leading spaces
Trailing spaces
Double spaces
Hidden characters
Tabs

Clean all instances.

Special Character Analysis

Find:

Symbols
Encoding errors
Corrupted characters
Unexpected punctuation

Correct where appropriate.

Date Quality Audit

Detect:

Invalid dates
Mixed date formats
Text dates
Impossible dates
Future dates (where inappropriate)

Standardize.

Numeric Quality Audit

Find:

Negative values
Outliers
Impossible values
Extreme values
Rounding inconsistencies

Assess business impact.

PHASE 4: SHEET-BY-SHEET ANALYSIS

For every sheet:

Generate:

Sheet Summary

Include:

Purpose
Row count
Column count
Key fields
Critical observations
Column Dictionary

For every column provide:

Column Meaning Data Type Quality Score

Create this automatically.

PHASE 5: ADVANCED DATA CLEANING

Apply intelligent cleaning.

Standardization

Standardize:

Dates
Names
Countries
States
Cities
Categories
Codes
Currency formats
Missing Value Treatment

Decide:

Remove
Replace
Impute
Leave unchanged

Provide rationale.

Duplicate Resolution

Keep:

Best record

Remove:

Redundant records

Document logic.

Data Type Corrections

Automatically correct:

Numbers stored as text
Dates stored as text
Percentage formats
Currency formats
PHASE 6: BUSINESS RULE DETECTION

Infer likely business rules.

Examples:

Customer Files

Detect:

Duplicate customer IDs
Invalid customer IDs
Missing contact data
Sales Files

Detect:

Quantity × Price ≠ Revenue
Missing sales values
Duplicate transactions
Inventory Files

Detect:

Negative inventory
Duplicate SKUs
Inventory mismatches
Finance Files

Detect:

Debit-credit imbalances
Calculation inconsistencies
Fiscal period errors
PHASE 7: CALCULATED FIELDS DISCOVERY

Determine whether additional columns should exist.

Examples:

Sales

Add:

Revenue
Margin
Profit
Discount %
HR

Add:

Tenure
Age
Experience
Finance

Add:

Variance
Growth %
Forecast

Generate useful calculated columns where appropriate.

PHASE 8: ANALYTICAL INSIGHTS

Generate business insights.

Identify:

Trends
Patterns
Anomalies
Outliers
Risks

Opportunities
Data Gaps
Reporting Improvements

Provide executive-level commentary.

PHASE 9: IMPROVEMENT RECOMMENDATIONS

Create prioritized recommendations.

Categorize:

Critical

Issues impacting accuracy.

High

Issues impacting reporting.

Medium

Quality improvements.

Low

Formatting improvements.

Estimate impact.

PHASE 10: OUTPUT FILE CREATION

For EVERY uploaded file:

Create a cleaned version.

Do not overwrite original data.

Create:

Sheet 1+

Original sheets cleaned.

New Sheet: DATA_QUALITY_REPORT

Include:

Category Issue Count Impact Action Taken
New Sheet: CHANGES_LOG

Track every modification.

Include:

Sheet Column Original Updated Reason
New Sheet: SUMMARY_DASHBOARD

Include:

Total rows
Total columns
Missing values fixed
Duplicates removed
Columns standardized
Data quality score
Overall health score
New Sheet: BUSINESS_INSIGHTS

Include:

Findings
Risks
Opportunities
Recommendations
QUALITY STANDARDS

Before delivering:

Verify:

■ No data corruption
■ No accidental deletions
■ Formulas preserved where applicable
■ Formatting consistency maintained
■ Audit trail created
■ All changes logged
■ Business logic validated
■ Data quality improved measurably

FINAL DELIVERABLE EXPECTATION

For each uploaded file:

Understand file purpose.
Audit every sheet.
Audit every column.
Detect all quality issues.
Clean and standardize data.
Add useful calculated fields where appropriate.
Generate insights.
Create DATA_QUALITY_REPORT sheet.
Create CHANGES_LOG sheet.
Create SUMMARY_DASHBOARD sheet.
Create BUSINESS_INSIGHTS sheet.
Provide a detailed executive summary describing all work performed, findings, risks, improvements, and recommendations.

The objective is not merely cleaning data, but transforming each file into a high-quality, analytics-ready, governance-compliant dataset with complete transparency of every change made.If you're testing the absolute limits of an AI's data engineering, data governance, analytics, auditing, and data quality capabilities, your prompt should behave like a formal Statement of Work (SOW), Data Governance Framework, Quality Assurance Specification, and Business Intelligence Engagement rolled into one.

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You are acting as:

Chief Data Officer (CDO)
Senior Data Architect
Data Governance Lead
Data Quality Specialist
Data Engineer
Business Analyst
Business Intelligence Consultant
Master Data Management Specialist
Data Steward
Data Auditor
Financial Analyst
Statistical Analyst
Reporting Specialist
Excel Expert
Quality Assurance Auditor

Your objective is NOT simply to clean data.

Your objective is to fully understand, assess, improve, validate, enrich, document, transform, optimize, and future-proof every uploaded file.

You must think and work like a consulting team conducting a multimillion-dollar enterprise data modernization project.

2. CORE PHILOSOPHY

Before changing anything:

You must understand:

What the file represents
Why the file exists
What business process created it
Who likely uses it
What decisions depend on it
What mistakes currently exist
What risks are hidden within it
What opportunities are being missed

Never perform blind cleaning.

Every change must be:

Justified
Documented
Traceable
Reversible
Auditable

Assume the file may be used for:

Executive reporting
Financial planning
Customer management
Regulatory compliance
Operations management
Inventory management
Strategic decision-making
3. FILE DISCOVERY PHASE

For every uploaded file create a complete discovery assessment.

3.1 File Overview

Capture:

File name
File extension
File size
Number of sheets
Number of rows
Number of columns
Number of formulas
Number of calculated fields
Number of hidden sheets
Number of hidden columns
Number of empty sheets
Number of protected sheets

Provide observations.

3.2 Dataset Classification

Determine if the file primarily contains:

Sales data
CRM data
Customer files

Financial records
Accounting data
Payroll records
Supply chain data
Inventory datasets
Manufacturing records
HR records
Marketing datasets
Project tracking
Forecast data
Budget information
Operational metrics

Explain reasoning.

4. BUSINESS UNDERSTANDING ENGINE

Reverse engineer the business purpose.

Determine:

What business function owns this data?

Possibilities:

Finance
Sales
Marketing
HR
Operations
Procurement
Logistics
Customer Support
Manufacturing
What business questions is this data intended to answer?

Examples:

Revenue performance
Customer retention
Profitability
Employee turnover
Stock management
Cash flow

Provide business interpretation.

5. DATA STRUCTURE ASSESSMENT

Evaluate structural quality.

Header Analysis

Detect:

Blank headers
Duplicated headers
Multi-row headers
Misaligned headers
Header spelling mistakes
Truncated labels
Inconsistent naming conventions

Recommend standardized naming framework.

Sheet Architecture Review

Evaluate:

Sheet organization
Naming standards

Navigation friendliness
Maintainability
Data flow structure
Formula dependencies

Score architecture quality from 1-100.

6. DATA QUALITY ASSESSMENT FRAMEWORK

Perform enterprise-grade profiling.

Missing Value Assessment

Identify every form of missing data:

Null
Blank
Empty string
Space-only values
N/A
NA
NULL
Missing
Unknown
Not Available
Dash (-)

For every column report:

Missing count
Missing percentage
Impact severity
Recommended action
Duplicate Analysis

Generate:

Exact Duplicate Review

Identify:

Fully duplicated records
Near Duplicate Review

Find:

Similar names
Similar addresses
Similar customers
Intelligent Duplicate Detection

Identify:

Potential same entity
Slight spelling differences
Case variations
Formatting variations

Generate confidence score.

Consistency Checks

Review:

Date formatting
Currency formatting
Category consistency
Naming standards
Country names
Product codes
Entity identifiers

Generate inconsistency report.

7. DATA TYPE VALIDATION ENGINE

For every column:

Determine:

Expected datatype
Actual datatype
Data mismatch percentage

Examples:

Numeric stored as text
Dates stored as text
Boolean stored as string
Currency stored as general format

Auto-correct where safe.

Document all corrections.

8. WHITESPACE AND CHARACTER AUDIT

Check for:

Leading spaces
Trailing spaces
Double spaces
Triple spaces
Non-breaking spaces
Tab characters
Hidden characters
Unicode issues

Clean all safely.

Document counts.

9. DATE INTELLIGENCE ANALYSIS

Inspect:

Invalid dates
Incomplete dates
Incorrect years
Impossible dates
Future dates
Historical anomalies

Normalize all valid dates.

Use a single date standard across the workbook.

10. NUMERICAL VALIDATION ENGINE

Evaluate:

Negative values
Zero values
Missing values
Extremely high values
Outliers
Suspicious patterns

Apply statistical profiling:

Mean
Median

Standard deviation
Variance
Quartiles
Percentile analysis

Flag anomalies.

11. COLUMN-BY-COLUMN DICTIONARY CREATION

Create a business dictionary.

For every column:

Include:

Column name
Business definition
Data type
Null percentage
Example values
Data quality score
Recommended usage

Generate automatically.

12. RELATIONSHIP ANALYSIS

Identify relationships between sheets.

Detect:

Primary keys
Foreign keys
Composite keys
Lookup relationships
Reference tables
Fact tables
Dimension tables

Build relationship map.

Identify broken relationships.

13. BUSINESS RULE DISCOVERY

Infer business logic automatically.

Examples:

Sales Files

Check:

Revenue = Quantity × Price

Discount ≤ 100%

Sales Date ≤ Current Date

Finance Files

Check:

Assets = Liabilities + Equity

Debit = Credit

Period totals reconcile

Inventory Files

Check:

Stock ≥ 0

Stock movements reconcile

SKU uniqueness

HR Files

Check:

Age validity

Joining date validity

Termination date logic

Create custom validations.

14. ADVANCED CLEANING ENGINE

Implement intelligent standardization.

Text Standardization

Normalize:

Names
Job titles
Product names
Customer names
Locations
Geographic Standardization

Standardize:

Countries
States
Regions
Cities
Categorical Standardization

Consolidate:

Examples:

USA U.S.A. United States

→ United States

Document every mapping.

15. SMART DATA ENRICHMENT

Where possible create useful additional fields.

Examples:

Sales

Add:

Revenue
Margin
Margin %
Growth %
Sales Category
Finance

Add:

Variance
Trend
Ratio Analysis
Growth Indicators
Customers

Add:

Customer Lifetime Value
Segment
Risk Category
Inventory

Add:

Inventory Age
Turnover Ratio
Stock Health
16. ANALYTICS AND INSIGHTS GENERATION

Generate executive-level findings.

Identify:

Trends
Patterns
Concentrations
Outliers
Risks
Opportunities
Operational inefficiencies
Data quality concerns
Reporting weaknesses
Forecasting opportunities

Provide business-friendly interpretation.

17. ROOT CAUSE ANALYSIS

For major data quality issues:

Determine likely causes.

Examples:

Manual entry errors
System migration issues
Integration failures
Inconsistent business processes
Legacy data structures

Explain reasoning.

18. DATA GOVERNANCE ASSESSMENT

Evaluate:

Accuracy
Completeness
Consistency
Validity
Timeliness
Uniqueness

Generate scores.

Overall Data Quality Score

Create weighted scores:

Dimension ScoreCompleteness %
Accuracy %
Validity %
Consistency %
Timeliness %
Uniqueness %

Generate final score.

19. DASHBOARD CREATION

Create a sheet named:

EXECUTIVE_DASHBOARD

Include:

Total Records
Total Columns
Missing Values Fixed
Duplicates Removed
Outliers Found
Formulas Repaired
Quality Score
Risk Score

Use visual formatting.

20. CHANGE MANAGEMENT LOG

Create sheet:

CHANGE_LOG

Document every modification.

Fields:

Timestamp
Sheet
Column
Original Value
Updated Value
Reason
Rule Applied

Nothing should be changed without logging.

21. DATA_QUALITY_REPORT SHEET

Create comprehensive report containing:

Issue Type
Severity
Count
Impact
Resolution
Recommendation
22. BUSINESS_INSIGHTS SHEET

Generate:

Key Findings
Trends
Risks
Opportunities
Recommendations
Quick Wins
Strategic Improvements

Written for executives.

23. FUTURE IMPROVEMENT RECOMMENDATIONS

Recommend:

Validation rules
Excel controls
Data entry improvements
Governance standards
Automation opportunities
Power BI opportunities
Dashboard opportunities
Data model enhancements
24. FINAL DELIVERABLE REQUIREMENTS

For EVERY uploaded file:

Deliver:

Cleaned Data
Standardized Data
Enhanced Data
Quality Report
Change Log
Executive Dashboard
Business Insights
Relationship Analysis
Data Dictionary
Governance Assessment
Recommendations
25. SUCCESS CRITERIA

The project is successful only if:

■ Every sheet has been reviewed
■ Every column has been analyzed
■ Every issue has been documented
■ Every change has been logged
■ Every correction is justified
■ Data quality score improves significantly
■ Data becomes analytics-ready
■ Reporting becomes more reliable
■ Business value is increased
■ Executive users can trust the output

FINAL INSTRUCTION TO THE AI

Do not behave like a spreadsheet cleaner.

Behave like an elite data consulting firm conducting a complete enterprise data audit, remediation, governance, business intelligence, and data modernization engagement.

Your goal is to transform every uploaded file into a trusted, production-ready, decision-grade data asset while maintaining full transparency, traceability, auditability, and business relevance of every action performed.If you're testing the absolute limits of an AI's data engineering, data governance, analytics, auditing, and data quality capabilities, your prompt should behave like a formal Statement of Work (SOW), Data Governance Framework, Quality Assurance Specification, and Business Intelligence Engagement rolled into one.

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Number of empty sheets
Number of protected sheets

Provide observations.

3.2 Dataset Classification

Determine if the file primarily contains:

Sales data
CRM data
Customer files
Financial records
Accounting data
Payroll records
Supply chain data
Inventory datasets
Manufacturing records
HR records
Marketing datasets
Project tracking
Forecast data
Budget information
Operational metrics

Explain reasoning.

4. BUSINESS UNDERSTANDING ENGINE

Reverse engineer the business purpose.

Determine:

What business function owns this data?

Possibilities:

Finance
Sales
Marketing
HR
Operations
Procurement
Logistics
Customer Support
Manufacturing
What business questions is this data intended to answer?

Examples:

Revenue performance
Customer retention
Profitability
Employee turnover

Stock management
Cash flow

Provide business interpretation.

5. DATA STRUCTURE ASSESSMENT

Evaluate structural quality.

Header Analysis

Detect:

Blank headers
Duplicated headers
Multi-row headers
Misaligned headers
Header spelling mistakes
Truncated labels
Inconsistent naming conventions

Recommend standardized naming framework.

Sheet Architecture Review

Evaluate:

Sheet organization
Naming standards
Navigation friendliness
Maintainability
Data flow structure
Formula dependencies

Score architecture quality from 1-100.

6. DATA QUALITY ASSESSMENT FRAMEWORK

Perform enterprise-grade profiling.

Missing Value Assessment

Identify every form of missing data:

Null
Blank
Empty string
Space-only values
N/A
NA
NULL
Missing
Unknown
Not Available
Dash (-)

For every column report:

Missing count
Missing percentage
Impact severity
Recommended action
Duplicate Analysis

Generate:

Exact Duplicate Review

Identify:

Fully duplicated records

Near Duplicate Review

Find:

Similar names
Similar addresses
Similar customers
Intelligent Duplicate Detection

Identify:

Potential same entity
Slight spelling differences
Case variations
Formatting variations

Generate confidence score.

Consistency Checks

Review:

Date formatting
Currency formatting
Category consistency
Naming standards
Country names
Product codes
Entity identifiers

Generate inconsistency report.

7. DATA TYPE VALIDATION ENGINE

For every column:

Determine:

Expected datatype
Actual datatype
Data mismatch percentage

Examples:

Numeric stored as text
Dates stored as text
Boolean stored as string
Currency stored as general format

Auto-correct where safe.

Document all corrections.

8. WHITESPACE AND CHARACTER AUDIT

Check for:

Leading spaces
Trailing spaces
Double spaces
Triple spaces
Non-breaking spaces
Tab characters
Hidden characters
Unicode issues

Clean all safely.

Document counts.

9. DATE INTELLIGENCE ANALYSIS

Inspect:

Invalid dates
Incomplete dates
Incorrect years
Impossible dates
Future dates
Historical anomalies

Normalize all valid dates.

Use a single date standard across the workbook.

10. NUMERICAL VALIDATION ENGINE

Evaluate:

Negative values
Zero values
Missing values
Extremely high values
Outliers
Suspicious patterns

Apply statistical profiling:

Mean
Median
Standard deviation
Variance
Quartiles
Percentile analysis

Flag anomalies.

11. COLUMN-BY-COLUMN DICTIONARY CREATION

Create a business dictionary.

For every column:

Include:

Column name
Business definition
Data type
Null percentage
Example values
Data quality score
Recommended usage

Generate automatically.

12. RELATIONSHIP ANALYSIS

Identify relationships between sheets.

Detect:

Primary keys
Foreign keys
Composite keys
Lookup relationships
Reference tables
Fact tables
Dimension tables

Build relationship map.

Identify broken relationships.

13. BUSINESS RULE DISCOVERY

Infer business logic automatically.

Examples:

Sales Files

Check:

Revenue = Quantity × Price

Discount ≤ 100%

Sales Date ≤ Current Date

Finance Files

Check:

Assets = Liabilities + Equity

Debit = Credit

Period totals reconcile

Inventory Files

Check:

Stock ≥ 0

Stock movements reconcile

SKU uniqueness

HR Files

Check:

Age validity

Joining date validity

Termination date logic

Create custom validations.

14. ADVANCED CLEANING ENGINE

Implement intelligent standardization.

Text Standardization

Normalize:

Names
Job titles
Product names
Customer names
Locations
Geographic Standardization

Standardize:

Countries
States
Regions
Cities
Categorical Standardization

Consolidate:

Examples:

USA U.S.A. United States

→ United States

Document every mapping.

15. SMART DATA ENRICHMENT

Where possible create useful additional fields.

Examples:

Sales

Add:

Revenue
Margin
Margin %
Growth %
Sales Category
Finance

Add:

Variance
Trend
Ratio Analysis
Growth Indicators
Customers

Add:

Customer Lifetime Value
Segment
Risk Category
Inventory

Add:

Inventory Age
Turnover Ratio
Stock Health
16. ANALYTICS AND INSIGHTS GENERATION

Generate executive-level findings.

Identify:

Trends
Patterns
Concentrations
Outliers
Risks
Opportunities
Operational inefficiencies
Data quality concerns
Reporting weaknesses
Forecasting opportunities

Provide business-friendly interpretation.

17. ROOT CAUSE ANALYSIS

For major data quality issues:

Determine likely causes.

Examples:

Manual entry errors
System migration issues
Integration failures
Inconsistent business processes
Legacy data structures

Explain reasoning.

18. DATA GOVERNANCE ASSESSMENT

Evaluate:

Accuracy
Completeness
Consistency
Validity
Timeliness
Uniqueness

Generate scores.

Overall Data Quality Score

Create weighted scores:

Dimension ScoreCompleteness %
Accuracy %
Validity %
Consistency %
Timeliness %
Uniqueness %

Generate final score.

19. DASHBOARD CREATION

Create a sheet named:

EXECUTIVE_DASHBOARD

Include:

Total Records
Total Columns
Missing Values Fixed
Duplicates Removed
Outliers Found
Formulas Repaired
Quality Score
Risk Score

Use visual formatting.

20. CHANGE MANAGEMENT LOG

Create sheet:

CHANGE_LOG

Document every modification.

Fields:

Timestamp
Sheet
Column
Original Value

Updated Value
Reason
Rule Applied

Nothing should be changed without logging.

21. DATA_QUALITY_REPORT SHEET

Create comprehensive report containing:

Issue Type
Severity
Count
Impact
Resolution
Recommendation
22. BUSINESS_INSIGHTS SHEET

Generate:

Key Findings
Trends
Risks
Opportunities
Recommendations
Quick Wins
Strategic Improvements

Written for executives.

23. FUTURE IMPROVEMENT RECOMMENDATIONS

Recommend:

Validation rules
Excel controls
Data entry improvements
Governance standards
Automation opportunities
Power BI opportunities
Dashboard opportunities
Data model enhancements
24. FINAL DELIVERABLE REQUIREMENTS

For EVERY uploaded file:

Deliver:

Cleaned Data
Standardized Data
Enhanced Data
Quality Report
Change Log
Executive Dashboard
Business Insights
Relationship Analysis
Data Dictionary
Governance Assessment
Recommendations
25. SUCCESS CRITERIA

The project is successful only if:

■ Every sheet has been reviewed
■ Every column has been analyzed
■ Every issue has been documented
■ Every change has been logged
■ Every correction is justified
■ Data quality score improves significantly
■ Data becomes analytics-ready
■ Reporting becomes more reliable

■ Business value is increased
■ Executive users can trust the output

FINAL INSTRUCTION TO THE AI

Do not behave like a spreadsheet cleaner.

Behave like an elite data consulting firm conducting a complete enterprise data audit, remediation, governance, business intelligence, and data modernization engagement.

Your goal is to transform every uploaded file into a trusted, production-ready, decision-grade data asset while maintaining full transparency, traceability, auditability, and business relevance of every action performed.Act as an elite enterprise data consulting firm comprising a Chief Data Officer, Senior Data Architect, Data Engineer, Data Governance Lead, Data Quality Auditor, Business Analyst, BI Consultant, Financial Analyst, Master Data Management Specialist, and Excel Expert. Your responsibility is not to merely clean uploaded files but to perform a comprehensive end-to-end data assessment, remediation, enrichment, governance review, and analytics transformation exercise. For every uploaded Excel, CSV, TXT, XLSX, or multi-sheet dataset, first understand the business context before making any modification. Reverse engineer what the data represents, identify its probable business use case, determine the department or function that owns the data, infer the key decisions supported by the dataset, and establish how the information may be used in reporting, forecasting, planning, compliance, operations, or executive decision-making. Never perform blind cleaning. Every change must be justified, documented, traceable, auditable, and reversible.

Begin by conducting a complete file discovery and profiling exercise. Analyze file metadata including filename, extension, file size, workbook structure, sheet count, row count, column count, formula count, calculated fields, hidden sheets, hidden rows, hidden columns, protected elements, merged cells, named ranges, pivot tables, charts, and workbook dependencies. Assess the overall architecture of the workbook and determine whether the structure is suitable for reporting, analytics, automation, or long-term maintenance. Evaluate sheet organization, navigation quality, naming conventions, and structural consistency. Generate a workbook health assessment and provide an architectural score between 0 and 100 with detailed reasoning.

Perform deep business understanding and semantic interpretation. Determine whether the dataset relates to sales, finance, accounting, customer management, CRM, procurement, supply chain, logistics, inventory, manufacturing, human resources, project management, marketing, operations, budgeting, forecasting, compliance, or executive reporting. Develop a written explanation describing the likely purpose of every sheet and explain how different sheets interact with one another. Identify key entities such as customers, products, employees, vendors, transactions, invoices, orders, projects, assets, inventory items, financial accounts, cost centers, departments, or business units. Create a business interpretation report that explains the dataset in plain language suitable for both technical and non-technical audiences.

Conduct a sheet-by-sheet and column-by-column forensic audit. Examine all headers and identify blank headers, duplicated headers, merged headers, malformed headers, misaligned columns, truncated names, inconsistent naming conventions, spelling mistakes, special characters, hidden text, and multi-row header structures. Propose standardized naming conventions and implement them where appropriate. Review column ordering, logical grouping, schema consistency, and structural integrity. Identify redundant columns, obsolete fields, duplicate columns, unused attributes, and potential normalization opportunities. Evaluate whether the workbook follows logical data modeling principles and document all observations.

Perform an enterprise-grade data quality assessment on every column and every record. Detect all forms of missing information including nulls, blanks, empty strings, spaces, tabs, N/A, NA, NULL, Unknown, Missing, Not Available, dashes, placeholders, invalid defaults, and system-generated blank values. Measure missing-value counts and percentages for every column. Rate the severity and business impact of each missing-data issue and recommend appropriate remediation actions. Evaluate whether data should be imputed, standardized, corrected, deleted, retained, or escalated for review. Provide detailed reasoning behind every recommendation.

Run comprehensive duplicate detection analysis. Identify exact duplicates, partial duplicates, fuzzy duplicates, near matches, entity duplicates, customer duplicates, vendor duplicates, employee duplicates, transaction duplicates, invoice duplicates, order duplicates, and record fragmentation scenarios. Use intelligent matching techniques including spelling similarity, formatting differences, case variations, spacing anomalies, abbreviation patterns, and identifier inconsistencies. Assign confidence levels to duplicate candidates and document removal or consolidation decisions in detail. Preserve evidence of duplicate detection in the audit report.

Validate all data types and compare actual values against expected formats. Detect numeric values stored as text, dates stored as text, currencies stored in inconsistent formats, percentages

represented incorrectly, invalid booleans, malformed identifiers, and mixed datatype columns. Correct data types wherever safe and document all transformations. Validate date structures thoroughly by identifying invalid dates, impossible dates, incorrect years, future dates where not expected, historical outliers, inconsistent regional formats, incomplete dates, text-based dates, and timestamp irregularities. Standardize all date values into a single enterprise-friendly format while preserving business meaning.

Perform whitespace, encoding, and character-level analysis. Detect leading spaces, trailing spaces, double spaces, multiple spaces, tab characters, hidden symbols, non-breaking spaces, encoding issues, corrupted characters, Unicode inconsistencies, invisible formatting artifacts, and invalid special characters. Clean and normalize all applicable values while maintaining original business intent. Generate detailed counts and summaries of all remediation activities performed.

Execute advanced numerical validation procedures. Assess every numeric field for statistical integrity using methods such as minimum, maximum, mean, median, standard deviation, variance, quartiles, percentiles, outlier detection, distribution analysis, and anomaly detection. Identify impossible values, suspicious patterns, negative values where not expected, zeros where invalid, unusually large quantities, rounding inconsistencies, currency conversion anomalies, and computational inconsistencies. Explain the probable causes and business implications of each identified anomaly.

Create a complete data dictionary for every workbook. For every column, generate business definitions, inferred business purpose, datatype classification, null percentage, uniqueness percentage, sample values, validation rules, quality assessment score, sensitivity classification, and recommended usage guidelines. Present the dictionary in a dedicated worksheet so that future users can understand the meaning and intended purpose of every field without external documentation.

Analyze relationships across sheets and datasets. Identify primary keys, surrogate keys, foreign keys, composite keys, lookup relationships, reference tables, fact tables, dimension tables, hierarchies, parent-child relationships, and potential data model structures. Validate referential integrity and identify broken relationships, orphan records, duplicate keys, missing matches, and inconsistent mappings. Build a relationship inventory and document all dependencies found within the workbook.

Discover and validate business rules automatically. Infer likely calculations, reporting logic, validation rules, operational constraints, financial controls, inventory balances, sales formulas, payroll logic, customer lifecycle rules, and compliance requirements. Test whether quantities multiplied by prices equal revenue, whether discounts remain within sensible ranges, whether accounting balances reconcile, whether inventory can become negative, whether employee dates follow logical sequences, and whether transaction timestamps reflect valid business activity. Flag any inconsistencies and propose corrective actions.

Perform intelligent data standardization and enrichment. Normalize names, product descriptions, titles, countries, regions, states, cities, categories, business units, account names, cost centers, and operational codes. Consolidate multiple representations of the same entity into approved standard forms. Where appropriate, create derived fields that enhance analytical value. Examples include revenue calculations, margin calculations, margin percentages, year-over-year growth, month-over-month growth, profitability indicators, customer lifetime value, customer segmentation, risk classifications, inventory aging, turnover ratios, tenure calculations, performance indicators, utilization rates, variance analysis, forecasting metrics, and trend indicators. Clearly distinguish between original and AI-generated fields.

Generate a comprehensive analytics and business intelligence assessment. Identify trends, patterns, correlations, seasonal behavior, growth opportunities, operational inefficiencies, concentration risks, performance drivers, exceptional performers, declining segments, unusual activities, emerging concerns, hidden opportunities, forecasting possibilities, and strategic recommendations. Provide observations suitable for executives, managers, analysts, and operational teams. Explain not just what is happening in the data but why it may be happening and what actions should be considered.

Conduct root cause analysis for all significant data quality problems. Explain whether issues likely originate from manual data entry, system migrations, process breakdowns, integrations, inconsistent governance practices, legacy systems, spreadsheet misuse, poor validation controls, inadequate master data management, or reporting process deficiencies. Quantify the potential impact on decision-making, analytics, reporting accuracy, and business performance.

Perform a formal data governance assessment. Evaluate the dataset across the dimensions of Completeness, Accuracy, Consistency, Validity, Integrity, Timeliness, Uniqueness, Conformity, Reliability, Traceability, and Auditability. Assign numerical scores to each dimension and calculate an overall Data Quality Index. Clearly explain how the score was calculated and identify

the primary drivers that influence the rating.

For every file processed, create additional sheets named EXECUTIVE_DASHBOARD, DATA_QUALITY_REPORT, CHANGE_LOG, DATA_DICTIONARY, RELATIONSHIP_MAP, BUSINESS_INSIGHTS, GOVERNANCE_ASSESSMENT, and RECOMMENDATIONS. The EXECUTIVE_DASHBOARD should provide a management-level overview including total records, total columns, missing values detected and fixed, duplicates removed, outliers identified, formulas repaired, data quality scores, governance maturity ratings, and major business observations. The DATA_QUALITY_REPORT should list all issues, severity levels, impact assessments, remediation actions, and improvement opportunities. The CHANGE_LOG must document every modification including sheet name, column name, original value, updated value, rule applied, date of change, and business justification. The BUSINESS_INSIGHTS sheet should contain key findings, trends, risks, opportunities, executive commentary, and high-priority recommendations. The DATA_DICTIONARY should fully document all fields. The GOVERNANCE_ASSESSMENT should summarize quality dimensions and scores. The RECOMMENDATIONS sheet should provide both short-term fixes and long-term transformation opportunities.

Finally, produce a detailed executive summary that explains the current state of the dataset, major risks identified, quality improvements completed, governance observations, analytical value discovered, business opportunities revealed, limitations encountered, assumptions made, and future recommendations. Treat this engagement as if the dataset will be used by senior leadership for critical decision-making. The ultimate objective is to transform every uploaded file from a potentially unreliable data source into a trusted, analytics-ready, governance-compliant, production-quality enterprise data asset while ensuring complete transparency, traceability, accountability, and auditability for every action taken. give the revised separate 10 excel files here

## Prompt #2 - Follow-up: Highlight N/A in Green + Requirements (2026-10-28)

what are the changes you have made and make sure give the requirments which you need from my need like in n/a what needs to be filled or highlight those in green and give in a separate sheet for each excel so i can see and give you for those cleaning from my end wha tneeds to be done

## Prompt #3 - PDF Archive Generation Request (2026-10-28) - Current Request

```
Create a downloadable PDF that contains ONLY the prompts, requests, instructions, enhancement
requests, corrections, feature requests, specifications, and task descriptions that I personally
provided throughout this chat history.

Do NOT include:
- Assistant responses
- Explanations
- Generated content unless it was part of my original prompt
- Analysis or interpretations
- Additional requirements not explicitly requested by me

Include ONLY my input messages.

Structure the PDF as follows:

1. Cover Page
- Title: "My Complete Prompt Archive"
- Subtitle: "User Submitted Prompts & Requests"

2. Table of Contents

3. Chronological Prompt Collection
- List every prompt exactly as submitted
- Preserve original wording wherever possible
- Include timestamps/dates when available
- Maintain chronological order

4. Prompt Iteration History
- Group related prompts together
- Show how requirements evolved over time
- Include follow-up enhancement requests

5. Application Development Requests
- All prompts related to application creation
- Feature additions
- Enhancement requests
- Missing functionality requests
- UI/UX requests
- Export and reporting requests

6. Document Generation Requests
- Prompts requesting specifications
- PDF creation requests
- Large documentation requests
- Expansion requests

7. Code Generation Requests
- Full application requests
- HTML/CSS/JS generation requests
- Single-file application requests
- Architecture and implementation requests

8. Feature Gap & Improvement Requests
- Missing feature observations
- Corrections requested
- Additional functionality requests
- Export requirements
- Data persistence requirements

9. Appendix
- Complete raw prompt log
- Unedited user messages only

Formatting Requirements:
- Professional PDF layout
- Clear section headings
- Page numbers
- Consistent typography
- One prompt per entry
- Preserve original intent and wording
```

- No assistant-generated content
- No summaries replacing prompts
- No omissions

Final Output:
Generate a clean, downloadable PDF containing every prompt and request I submitted in this conversation history, organized and structured, while preserving the original wording and chronological flow.

# 4. Prompt Iteration History

## Evolution of Requirements:

Phase 1: Initial comprehensive enterprise-grade data audit, cleaning, standardization, enrichment, governance framework for 10 Excel files (02-10 provided). Request included 25 phases covering file discovery, business understanding, data quality, dictionary, relationship analysis, business rules, enrichment, analytics, governance scoring, and 8 output sheets per file.

Phase 2: User observed gaps - requested explicit change log, requirements that need user input (N/A handling), green highlighting for missing values, and separate sheet per Excel for user to fill. This is a feature gap / improvement request building on Phase 1.

Phase 3: User requested archive of all prompts provided throughout chat, structured into 9 sections with professional PDF layout, containing ONLY user input messages.

# 5. Application Development Requests

## Related to application creation, feature additions, UI/UX, export/reporting:

- Prompt #1: Create cleaned versions for EVERY uploaded file with sheets: DATA_QUALITY_REPORT, CHANGES_LOG, SUMMARY_DASHBOARD / EXECUTIVE_DASHBOARD, BUSINESS_INSIGHTS, DATA_DICTIONARY, RELATIONSHIP_MAP, GOVERNANCE_ASSESSMENT, RECOMMENDATIONS. Deliver revised separate 10 excel files.

- Prompt #2: Highlight N/A / missing values in green and give in a separate sheet for each excel so user can fill and return for cleaning.

# 6. Document Generation Requests

- Prompt #1 includes request for detailed executive summary describing all work performed, findings, risks, improvements, recommendations. Also requires documentation sheets as deliverables.

- Prompt #3: Create downloadable PDF that contains ONLY prompts, requests, instructions, enhancement requests, corrections, feature requests, specifications, task descriptions personally provided. Structure: Cover Page, TOC, Chronological Collection, Iteration History, Application Development Requests, Document Generation Requests, Code Generation Requests, Feature Gap & Improvement Requests, Appendix. Formatting: Professional layout, page numbers, consistent typography, one prompt per entry, preserve original wording.

# 7. Code Generation Requests

- Prompt #1 implicitly requests data engineering code implementation to produce cleaned Excel files with formulas, calculated fields (Revenue, Margin, Tenure, Age, Variance, Growth %, Customer Lifetime Value, Inventory Age, Turnover Ratio etc.), standardization logic, validation rules.

## 8. Feature Gap & Improvement Requests

- Prompt #2: 'what are the changes you have made and make sure give the requirments which you need from my need like in n/a what needs to be filled or highlight those in green and give in a separate sheet for each excel so i can see and give you for those cleaning from my end wha tneeds to be done' - This is explicit missing feature observation: need transparency on changes and actionable list of fields requiring user input, with green highlighting.

# 9. Appendix - Complete Raw Prompt Log (Unedited User Messages Only)

## User Message 1 (Full Text):

```
Below is a comprehensive "10-page-level" master prompt that you can use with Copilot, ChatGPT,
Claude, Gemini, or any advanced AI data-cleaning assistant whenever you upload Excel, CSV, TXT, or
mixed data files.

MASTER PROMPT: ADVANCED DATA AUDIT, CLEANING, STANDARDIZATION & ANALYTICS ENGINE
ROLE

Act as a Senior Data Quality Consultant, Data Engineer, Data Analyst, Data Governance Specialist,
and Business Intelligence Auditor simultaneously.

You are not just cleaning files.

You must first understand:

What the file contains
Business purpose of the file
Column meanings
Relationships between sheets
Data quality issues
Missing business logic
Missing calculations
Structural flaws
Analytical opportunities
Reporting opportunities
Risks and anomalies

Before making any changes, deeply inspect and understand the complete file.

PHASE 1: FILE UNDERSTANDING

For every uploaded file:

Perform a complete audit.

Identify:

File Metadata
File type
Number of sheets
Number of rows
Number of columns
File size
Data types
Business Understanding

Determine:

What the file appears to represent
Domain/industry
Purpose of data
Intended use
Reporting purpose
Key business entities

Examples:

Sales Data
Customer Data
Financial Records
Inventory Data
HR Data
Logistics Data
Operations Data
Marketing Data
```

Provide a business interpretation.

PHASE 2: STRUCTURAL AUDIT

Audit every sheet independently.

Identify:

Header Issues
Missing headers
Duplicate headers
Blank headers
Merged headers
Misaligned headers
Multi-row headers
Column Issues
Duplicate columns
Redundant columns
Unused columns
Hidden columns
Misnamed columns
Sheet Issues
Empty sheets
Duplicate sheets
Hidden sheets
Broken references

Document all findings.

PHASE 3: DATA QUALITY ANALYSIS

Inspect every column individually.

Identify:

Missing Data

Check for:

Null values
Blank values
Empty strings
NA
N/A
NULL
Unknown
Missing

Provide:

Count
Percentage
Impact
Duplicate Analysis

Identify:

Exact duplicates
Partial duplicates
Near duplicates
Customer duplicates
Product duplicates
Transaction duplicates

Provide recommendations.

Data Type Validation

Verify:

Numbers

Dates
Currency
Text
Percentages
IDs

Find:

Mixed formats
Incorrect formats
Conversion issues
Case Consistency

Detect:

UPPER CASE
lower case
Title Case
Mixed Case

Standardize appropriately.

Whitespace Analysis

Identify:

Leading spaces
Trailing spaces
Double spaces
Hidden characters
Tabs

Clean all instances.

Special Character Analysis

Find:

Symbols
Encoding errors
Corrupted characters
Unexpected punctuation

Correct where appropriate.

Date Quality Audit

Detect:

Invalid dates
Mixed date formats
Text dates
Impossible dates
Future dates (where inappropriate)

Standardize.

Numeric Quality Audit

Find:

Negative values
Outliers
Impossible values
Extreme values
Rounding inconsistencies

Assess business impact.

PHASE 4: SHEET-BY-SHEET ANALYSIS

For every sheet:

Generate:

Sheet Summary

Include:

Purpose
Row count
Column count
Key fields
Critical observations
Column Dictionary

For every column provide:

Column Meaning Data Type Quality Score

Create this automatically.

PHASE 5: ADVANCED DATA CLEANING

Apply intelligent cleaning.

Standardization

Standardize:

Dates
Names
Countries
States
Cities
Categories
Codes
Currency formats
Missing Value Treatment

Decide:

Remove
Replace
Impute
Leave unchanged

Provide rationale.

Duplicate Resolution

Keep:

Best record

Remove:

Redundant records

Document logic.

Data Type Corrections

Automatically correct:

Numbers stored as text
Dates stored as text
Percentage formats
Currency formats
PHASE 6: BUSINESS RULE DETECTION

Infer likely business rules.

Examples:

Customer Files

Detect:

Duplicate customer IDs
Invalid customer IDs
Missing contact data
Sales Files

Detect:

Quantity × Price ≠ Revenue
Missing sales values
Duplicate transactions
Inventory Files

Detect:

Negative inventory
Duplicate SKUs
Inventory mismatches
Finance Files

Detect:

Debit-credit imbalances
Calculation inconsistencies
Fiscal period errors
PHASE 7: CALCULATED FIELDS DISCOVERY

Determine whether additional columns should exist.

Examples:

Sales

Add:

Revenue
Margin
Profit
Discount %
HR

Add:

Tenure
Age
Experience
Finance

Add:

Variance
Growth %
Forecast

Generate useful calculated columns where appropriate.

PHASE 8: ANALYTICAL INSIGHTS

Generate business insights.

Identify:

Trends
Patterns
Anomalies

Outliers
Risks
Opportunities
Data Gaps
Reporting Improvements

Provide executive-level commentary.

PHASE 9: IMPROVEMENT RECOMMENDATIONS

Create prioritized recommendations.

Categorize:

Critical

Issues impacting accuracy.

High

Issues impacting reporting.

Medium

Quality improvements.

Low

Formatting improvements.

Estimate impact.

PHASE 10: OUTPUT FILE CREATION

For EVERY uploaded file:

Create a cleaned version.

Do not overwrite original data.

Create:

Sheet 1+

Original sheets cleaned.

New Sheet: DATA_QUALITY_REPORT

Include:

Category Issue Count Impact Action Taken
New Sheet: CHANGES_LOG

Track every modification.

Include:

Sheet Column Original Updated Reason
New Sheet: SUMMARY_DASHBOARD

Include:

Total rows
Total columns
Missing values fixed
Duplicates removed
Columns standardized
Data quality score
Overall health score
New Sheet: BUSINESS_INSIGHTS

Include:

Findings
Risks
Opportunities
Recommendations
QUALITY STANDARDS

Before delivering:

Verify:

■ No data corruption
■ No accidental deletions
■ Formulas preserved where applicable
■ Formatting consistency maintained
■ Audit trail created
■ All changes logged
■ Business logic validated
■ Data quality improved measurably

FINAL DELIVERABLE EXPECTATION

For each uploaded file:

Understand file purpose.
Audit every sheet.
Audit every column.
Detect all quality issues.
Clean and standardize data.
Add useful calculated fields where appropriate.
Generate insights.
Create DATA_QUALITY_REPORT sheet.
Create CHANGES_LOG sheet.
Create SUMMARY_DASHBOARD sheet.
Create BUSINESS_INSIGHTS sheet.
Provide a detailed executive summary describing all work performed, findings, risks, improvements, and recommendations.

The objective is not merely cleaning data, but transforming each file into a high-quality, analytics-ready, governance-compliant dataset with complete transparency of every change made.If you're testing the absolute limits of an AI's data engineering, data governance, analytics, auditing, and data quality capabilities, your prompt should behave like a formal Statement of Work (SOW), Data Governance Framework, Quality Assurance Specification, and Business Intelligence Engagement rolled into one.

Below is a significantly expanded "enterprise-grade" version intended to drive extremely deep analysis across Excel, CSV, XLSX, TXT, and multi-sheet datasets.

ULTRA ADVANCED MASTER PROMPT
Enterprise Data Audit, Cleansing, Standardization, Enrichment, Modeling, Validation, Analytics & Governance Framework
1. ROLE AND RESPONSIBILITIES

You are acting as:

Chief Data Officer (CDO)
Senior Data Architect
Data Governance Lead
Data Quality Specialist
Data Engineer
Business Analyst
Business Intelligence Consultant
Master Data Management Specialist
Data Steward
Data Auditor
Financial Analyst
Statistical Analyst
Reporting Specialist
Excel Expert
Quality Assurance Auditor

Your objective is NOT simply to clean data.

Your objective is to fully understand, assess, improve, validate, enrich, document, transform, optimize, and future-proof every uploaded file.

You must think and work like a consulting team conducting a multimillion-dollar enterprise data modernization project.

2. CORE PHILOSOPHY

Before changing anything:

You must understand:

What the file represents
Why the file exists
What business process created it
Who likely uses it
What decisions depend on it
What mistakes currently exist
What risks are hidden within it
What opportunities are being missed

Never perform blind cleaning.

Every change must be:

Justified
Documented
Traceable
Reversible
Auditable

Assume the file may be used for:

Executive reporting
Financial planning
Customer management
Regulatory compliance
Operations management
Inventory management
Strategic decision-making
3. FILE DISCOVERY PHASE

For every uploaded file create a complete discovery assessment.

3.1 File Overview

Capture:

File name
File extension
File size
Number of sheets
Number of rows
Number of columns
Number of formulas
Number of calculated fields
Number of hidden sheets
Number of hidden columns
Number of empty sheets
Number of protected sheets

Provide observations.

3.2 Dataset Classification

Determine if the file primarily contains:

Sales data

CRM data
Customer files
Financial records
Accounting data
Payroll records
Supply chain data
Inventory datasets
Manufacturing records
HR records
Marketing datasets
Project tracking
Forecast data
Budget information
Operational metrics

Explain reasoning.

4. BUSINESS UNDERSTANDING ENGINE

Reverse engineer the business purpose.

Determine:

What business function owns this data?

Possibilities:

Finance
Sales
Marketing
HR
Operations
Procurement
Logistics
Customer Support
Manufacturing
What business questions is this data intended to answer?

Examples:

Revenue performance
Customer retention
Profitability
Employee turnover
Stock management
Cash flow

Provide business interpretation.

5. DATA STRUCTURE ASSESSMENT

Evaluate structural quality.

Header Analysis

Detect:

Blank headers
Duplicated headers
Multi-row headers
Misaligned headers
Header spelling mistakes
Truncated labels
Inconsistent naming conventions

Recommend standardized naming framework.

Sheet Architecture Review

Evaluate:

Sheet organization
Naming standards
Navigation friendliness
Maintainability
Data flow structure
Formula dependencies

Score architecture quality from 1-100.

6. DATA QUALITY ASSESSMENT FRAMEWORK

Perform enterprise-grade profiling.

Missing Value Assessment

Identify every form of missing data:

Null
Blank
Empty string
Space-only values
N/A
NA
NULL
Missing
Unknown
Not Available
Dash (-)

For every column report:

Missing count
Missing percentage
Impact severity
Recommended action
Duplicate Analysis

Generate:

Exact Duplicate Review

Identify:

Fully duplicated records
Near Duplicate Review

Find:

Similar names
Similar addresses
Similar customers
Intelligent Duplicate Detection

Identify:

Potential same entity
Slight spelling differences
Case variations
Formatting variations

Generate confidence score.

Consistency Checks

Review:

Date formatting
Currency formatting
Category consistency
Naming standards
Country names

Product codes
Entity identifiers

Generate inconsistency report.

7. DATA TYPE VALIDATION ENGINE

For every column:

Determine:

Expected datatype
Actual datatype
Data mismatch percentage

Examples:

Numeric stored as text
Dates stored as text
Boolean stored as string
Currency stored as general format

Auto-correct where safe.

Document all corrections.

8. WHITESPACE AND CHARACTER AUDIT

Check for:

Leading spaces
Trailing spaces
Double spaces
Triple spaces
Non-breaking spaces
Tab characters
Hidden characters
Unicode issues

Clean all safely.

Document counts.

9. DATE INTELLIGENCE ANALYSIS

Inspect:

Invalid dates
Incomplete dates
Incorrect years
Impossible dates
Future dates
Historical anomalies

Normalize all valid dates.

Use a single date standard across the workbook.

10. NUMERICAL VALIDATION ENGINE

Evaluate:

Negative values
Zero values
Missing values
Extremely high values
Outliers
Suspicious patterns

Apply statistical profiling:

Mean
Median
Standard deviation
Variance
Quartiles
Percentile analysis

Flag anomalies.

11. COLUMN-BY-COLUMN DICTIONARY CREATION

Create a business dictionary.

For every column:

Include:

Column name
Business definition
Data type
Null percentage
Example values
Data quality score
Recommended usage

Generate automatically.

12. RELATIONSHIP ANALYSIS

Identify relationships between sheets.

Detect:

Primary keys
Foreign keys
Composite keys
Lookup relationships
Reference tables
Fact tables
Dimension tables

Build relationship map.

Identify broken relationships.

13. BUSINESS RULE DISCOVERY

Infer business logic automatically.

Examples:

Sales Files

Check:

Revenue = Quantity × Price

Discount ≤ 100%

Sales Date ≤ Current Date

Finance Files

Check:

Assets = Liabilities + Equity

Debit = Credit

Period totals reconcile

Inventory Files

Check:

Stock ≥ 0

Stock movements reconcile

SKU uniqueness

HR Files

Check:

Age validity

Joining date validity

Termination date logic

Create custom validations.

14. ADVANCED CLEANING ENGINE

Implement intelligent standardization.

Text Standardization

Normalize:

Names
Job titles
Product names
Customer names
Locations
Geographic Standardization

Standardize:

Countries
States
Regions
Cities
Categorical Standardization

Consolidate:

Examples:

USA U.S.A. United States

→ United States

Document every mapping.

15. SMART DATA ENRICHMENT

Where possible create useful additional fields.

Examples:

Sales

Add:

Revenue
Margin
Margin %
Growth %
Sales Category
Finance

Add:

Variance
Trend
Ratio Analysis
Growth Indicators
Customers

Add:

Customer Lifetime Value
Segment
Risk Category
Inventory

Add:

Inventory Age
Turnover Ratio
Stock Health
16. ANALYTICS AND INSIGHTS GENERATION

Generate executive-level findings.

Identify:

Trends
Patterns
Concentrations
Outliers
Risks
Opportunities
Operational inefficiencies
Data quality concerns
Reporting weaknesses
Forecasting opportunities

Provide business-friendly interpretation.

17. ROOT CAUSE ANALYSIS

For major data quality issues:

Determine likely causes.

Examples:

Manual entry errors
System migration issues
Integration failures
Inconsistent business processes
Legacy data structures

Explain reasoning.

18. DATA GOVERNANCE ASSESSMENT

Evaluate:

Accuracy
Completeness
Consistency
Validity
Timeliness
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Generate scores.

Overall Data Quality Score

Create weighted scores:

Dimension ScoreCompleteness %
Accuracy %
Validity %
Consistency %
Timeliness %
Uniqueness %

Generate final score.

19. DASHBOARD CREATION

Create a sheet named:

EXECUTIVE_DASHBOARD

Include:

Total Records
Total Columns
Missing Values Fixed
Duplicates Removed
Outliers Found
Formulas Repaired
Quality Score
Risk Score

Use visual formatting.

20. CHANGE MANAGEMENT LOG

Create sheet:

CHANGE_LOG

Document every modification.

Fields:

Timestamp
Sheet
Column
Original Value
Updated Value
Reason
Rule Applied

Nothing should be changed without logging.

21. DATA_QUALITY_REPORT SHEET

Create comprehensive report containing:

Issue Type
Severity
Count
Impact
Resolution
Recommendation
22. BUSINESS_INSIGHTS SHEET

Generate:

Key Findings
Trends
Risks
Opportunities
Recommendations
Quick Wins
Strategic Improvements

Written for executives.

23. FUTURE IMPROVEMENT RECOMMENDATIONS

Recommend:

Validation rules
Excel controls
Data entry improvements
Governance standards
Automation opportunities
Power BI opportunities
Dashboard opportunities
Data model enhancements
24. FINAL DELIVERABLE REQUIREMENTS

For EVERY uploaded file:

Deliver:

Cleaned Data
Standardized Data
Enhanced Data
Quality Report
Change Log
Executive Dashboard
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Relationship Analysis
Data Dictionary
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25. SUCCESS CRITERIA

The project is successful only if:

■ Every sheet has been reviewed
■ Every column has been analyzed
■ Every issue has been documented
■ Every change has been logged
■ Every correction is justified
■ Data quality score improves significantly
■ Data becomes analytics-ready
■ Reporting becomes more reliable
■ Business value is increased
■ Executive users can trust the output

FINAL INSTRUCTION TO THE AI

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You are acting as:

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Senior Data Architect
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Data Quality Specialist
Data Engineer
Business Analyst
Business Intelligence Consultant
Master Data Management Specialist
Data Steward
Data Auditor
Financial Analyst
Statistical Analyst
Reporting Specialist
Excel Expert
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Your objective is to fully understand, assess, improve, validate, enrich, document, transform, optimize, and future-proof every uploaded file.

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2. CORE PHILOSOPHY

Before changing anything:

You must understand:

What the file represents
Why the file exists
What business process created it
Who likely uses it
What decisions depend on it
What mistakes currently exist
What risks are hidden within it
What opportunities are being missed

Never perform blind cleaning.

Every change must be:

Justified
Documented
Traceable
Reversible
Auditable

Assume the file may be used for:

Executive reporting
Financial planning
Customer management
Regulatory compliance
Operations management
Inventory management
Strategic decision-making
3. FILE DISCOVERY PHASE

For every uploaded file create a complete discovery assessment.

3.1 File Overview

Capture:

File name
File extension
File size
Number of sheets
Number of rows

Number of columns
Number of formulas
Number of calculated fields
Number of hidden sheets
Number of hidden columns
Number of empty sheets
Number of protected sheets

Provide observations.

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Determine if the file primarily contains:

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Evaluate structural quality.

Header Analysis

Detect:

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Recommend standardized naming framework.

Sheet Architecture Review

Evaluate:

Sheet organization
Naming standards
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Score architecture quality from 1-100.

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Identify:

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Find:

Similar names
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Similar customers
Intelligent Duplicate Detection

Identify:

Potential same entity

Slight spelling differences
Case variations
Formatting variations

Generate confidence score.

Consistency Checks

Review:

Date formatting
Currency formatting
Category consistency
Naming standards
Country names
Product codes
Entity identifiers

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Explain reasoning.

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Fields:

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Your goal is to transform every uploaded file into a trusted, production-ready, decision-grade data asset while maintaining full transparency, traceability, auditability, and business relevance of every action performed.Act as an elite enterprise data consulting firm comprising a Chief Data Officer, Senior Data Architect, Data Engineer, Data Governance Lead, Data Quality Auditor, Business Analyst, BI Consultant, Financial Analyst, Master Data Management Specialist, and Excel Expert. Your responsibility is not to merely clean uploaded files but to perform a comprehensive end-to-end data assessment, remediation, enrichment, governance review, and analytics transformation exercise. For every uploaded Excel, CSV, TXT, XLSX, or multi-sheet dataset, first understand the business context before making any modification. Reverse engineer what the data represents, identify its probable business use case, determine the department or function that owns the data, infer the key decisions supported by the dataset, and establish how the information may be used in reporting, forecasting, planning, compliance, operations, or executive decision-making. Never perform blind cleaning. Every change must be justified, documented, traceable, auditable, and reversible.

Begin by conducting a complete file discovery and profiling exercise. Analyze file metadata including filename, extension, file size, workbook structure, sheet count, row count, column count, formula count, calculated fields, hidden sheets, hidden rows, hidden columns, protected elements, merged cells, named ranges, pivot tables, charts, and workbook dependencies. Assess the overall architecture of the workbook and determine whether the structure is suitable for reporting, analytics, automation, or long-term maintenance. Evaluate sheet organization, navigation quality, naming conventions, and structural consistency. Generate a workbook health assessment and provide an architectural score between 0 and 100 with detailed reasoning.

Perform deep business understanding and semantic interpretation. Determine whether the dataset relates to sales, finance, accounting, customer management, CRM, procurement, supply chain, logistics, inventory, manufacturing, human resources, project management, marketing, operations, budgeting, forecasting, compliance, or executive reporting. Develop a written explanation describing the likely purpose of every sheet and explain how different sheets interact with one another. Identify key entities such as customers, products, employees, vendors, transactions, invoices, orders, projects, assets, inventory items, financial accounts, cost centers, departments, or business units. Create a business interpretation report that explains the dataset in plain language suitable for both technical and non-technical audiences.

Conduct a sheet-by-sheet and column-by-column forensic audit. Examine all headers and identify blank headers, duplicated headers, merged headers, malformed headers, misaligned columns, truncated names, inconsistent naming conventions, spelling mistakes, special characters, hidden text, and multi-row header structures. Propose standardized naming conventions and implement them where appropriate. Review column ordering, logical grouping, schema consistency, and structural integrity. Identify redundant columns, obsolete fields, duplicate columns, unused attributes, and potential normalization opportunities. Evaluate whether the workbook follows logical data modeling principles and document all observations.

Perform an enterprise-grade data quality assessment on every column and every record. Detect all forms of missing information including nulls, blanks, empty strings, spaces, tabs, N/A, NA, NULL, Unknown, Missing, Not Available, dashes, placeholders, invalid defaults, and system-generated blank values. Measure missing-value counts and percentages for every column. Rate the severity and business impact of each missing-data issue and recommend appropriate remediation actions. Evaluate whether data should be imputed, standardized, corrected, deleted, retained, or escalated for review. Provide detailed reasoning behind every recommendation.

Run comprehensive duplicate detection analysis. Identify exact duplicates, partial duplicates, fuzzy duplicates, near matches, entity duplicates, customer duplicates, vendor duplicates, employee duplicates, transaction duplicates, invoice duplicates, order duplicates, and record fragmentation scenarios. Use intelligent matching techniques including spelling similarity, formatting differences, case variations, spacing anomalies, abbreviation patterns, and identifier inconsistencies. Assign confidence levels to duplicate candidates and document removal or consolidation decisions in detail. Preserve evidence of duplicate detection in the audit report.

Validate all data types and compare actual values against expected formats. Detect numeric values stored as text, dates stored as text, currencies stored in inconsistent formats, percentages represented incorrectly, invalid booleans, malformed identifiers, and mixed datatype columns. Correct data types wherever safe and document all transformations. Validate date structures thoroughly by identifying invalid dates, impossible dates, incorrect years, future dates where not expected, historical outliers, inconsistent regional formats, incomplete dates, text-based dates, and timestamp irregularities. Standardize all date values into a single enterprise-friendly format while preserving business meaning.

Perform whitespace, encoding, and character-level analysis. Detect leading spaces, trailing spaces, double spaces, multiple spaces, tab characters, hidden symbols, non-breaking spaces, encoding issues, corrupted characters, Unicode inconsistencies, invisible formatting artifacts, and invalid special characters. Clean and normalize all applicable values while maintaining original business intent. Generate detailed counts and summaries of all remediation activities performed.

Execute advanced numerical validation procedures. Assess every numeric field for statistical integrity using methods such as minimum, maximum, mean, median, standard deviation, variance, quartiles, percentiles, outlier detection, distribution analysis, and anomaly detection. Identify impossible values, suspicious patterns, negative values where not expected, zeros where invalid, unusually large quantities, rounding inconsistencies, currency conversion anomalies, and computational inconsistencies. Explain the probable causes and business implications of each identified anomaly.

Create a complete data dictionary for every workbook. For every column, generate business definitions, inferred business purpose, datatype classification, null percentage, uniqueness percentage, sample values, validation rules, quality assessment score, sensitivity classification, and recommended usage guidelines. Present the dictionary in a dedicated worksheet so that future users can understand the meaning and intended purpose of every field without external documentation.

Analyze relationships across sheets and datasets. Identify primary keys, surrogate keys, foreign keys, composite keys, lookup relationships, reference tables, fact tables, dimension tables, hierarchies, parent-child relationships, and potential data model structures. Validate referential integrity and identify broken relationships, orphan records, duplicate keys, missing matches, and inconsistent mappings. Build a relationship inventory and document all dependencies found within the workbook.

Discover and validate business rules automatically. Infer likely calculations, reporting logic, validation rules, operational constraints, financial controls, inventory balances, sales formulas, payroll logic, customer lifecycle rules, and compliance requirements. Test whether quantities multiplied by prices equal revenue, whether discounts remain within sensible ranges, whether accounting balances reconcile, whether inventory can become negative, whether employee dates follow logical sequences, and whether transaction timestamps reflect valid business activity. Flag any inconsistencies and propose corrective actions.

Perform intelligent data standardization and enrichment. Normalize names, product descriptions, titles, countries, regions, states, cities, categories, business units, account names, cost centers, and operational codes. Consolidate multiple representations of the same entity into approved standard forms. Where appropriate, create derived fields that enhance analytical value. Examples include revenue calculations, margin calculations, margin percentages, year-over-year growth, month-over-month growth, profitability indicators, customer lifetime value, customer segmentation, risk classifications, inventory aging, turnover ratios, tenure calculations, performance indicators, utilization rates, variance analysis, forecasting metrics, and trend indicators. Clearly distinguish between original and AI-generated fields.

Generate a comprehensive analytics and business intelligence assessment. Identify trends, patterns, correlations, seasonal behavior, growth opportunities, operational inefficiencies, concentration risks, performance drivers, exceptional performers, declining segments, unusual activities, emerging concerns, hidden opportunities, forecasting possibilities, and strategic recommendations. Provide observations suitable for executives, managers, analysts, and operational teams. Explain not just what is happening in the data but why it may be happening and what actions should be considered.

Conduct root cause analysis for all significant data quality problems. Explain whether issues likely originate from manual data entry, system migrations, process breakdowns, integrations, inconsistent governance practices, legacy systems, spreadsheet misuse, poor validation controls, inadequate master data management, or reporting process deficiencies. Quantify the potential impact on decision-making, analytics, reporting accuracy, and business performance.

Perform a formal data governance assessment. Evaluate the dataset across the dimensions of

Completeness, Accuracy, Consistency, Validity, Integrity, Timeliness, Uniqueness, Conformity, Reliability, Traceability, and Auditability. Assign numerical scores to each dimension and calculate an overall Data Quality Index. Clearly explain how the score was calculated and identify the primary drivers that influence the rating.

For every file processed, create additional sheets named EXECUTIVE_DASHBOARD, DATA_QUALITY_REPORT, CHANGE_LOG, DATA_DICTIONARY, RELATIONSHIP_MAP, BUSINESS_INSIGHTS, GOVERNANCE_ASSESSMENT, and RECOMMENDATIONS. The EXECUTIVE_DASHBOARD should provide a management-level overview including total records, total columns, missing values detected and fixed, duplicates removed, outliers identified, formulas repaired, data quality scores, governance maturity ratings, and major business observations. The DATA_QUALITY_REPORT should list all issues, severity levels, impact assessments, remediation actions, and improvement opportunities. The CHANGE_LOG must document every modification including sheet name, column name, original value, updated value, rule applied, date of change, and business justification. The BUSINESS_INSIGHTS sheet should contain key findings, trends, risks, opportunities, executive commentary, and high-priority recommendations. The DATA_DICTIONARY should fully document all fields. The GOVERNANCE_ASSESSMENT should summarize quality dimensions and scores. The RECOMMENDATIONS sheet should provide both short-term fixes and long-term transformation opportunities.

Finally, produce a detailed executive summary that explains the current state of the dataset, major risks identified, quality improvements completed, governance observations, analytical value discovered, business opportunities revealed, limitations encountered, assumptions made, and future recommendations. Treat this engagement as if the dataset will be used by senior leadership for critical decision-making. The ultimate objective is to transform every uploaded file from a potentially unreliable data source into a trusted, analytics-ready, governance-compliant, production-quality enterprise data asset while ensuring complete transparency, traceability, accountability, and auditability for every action taken. give the revised separate 10 excel files here

## User Message 2 (Full Text):

what are the changes you have made and make sure give the requirments which you need from my need like in n/a what needs to be filled or highlight those in green and give in a separate sheet for each excel so i can see and give you for those cleaning from my end wha tneeds to be done

## User Message 3 (Full Text):

Create a downloadable PDF that contains ONLY the prompts, requests, instructions, enhancement requests, corrections, feature requests, specifications, and task descriptions that I personally provided throughout this chat history.

Do NOT include:
- Assistant responses
- Explanations
- Generated content unless it was part of my original prompt
- Analysis or interpretations
- Additional requirements not explicitly requested by me

Include ONLY my input messages.

Structure the PDF as follows:

1. Cover Page
- Title: "My Complete Prompt Archive"
- Subtitle: "User Submitted Prompts & Requests"

2. Table of Contents

3. Chronological Prompt Collection
- List every prompt exactly as submitted
- Preserve original wording wherever possible
- Include timestamps/dates when available
- Maintain chronological order

4. Prompt Iteration History
- Group related prompts together
- Show how requirements evolved over time
- Include follow-up enhancement requests

5. Application Development Requests
- All prompts related to application creation
- Feature additions
- Enhancement requests
- Missing functionality requests
- UI/UX requests
- Export and reporting requests

6. Document Generation Requests
- Prompts requesting specifications
- PDF creation requests
- Large documentation requests
- Expansion requests

7. Code Generation Requests
- Full application requests
- HTML/CSS/JS generation requests
- Single-file application requests
- Architecture and implementation requests

8. Feature Gap & Improvement Requests
- Missing feature observations
- Corrections requested
- Additional functionality requests
- Export requirements
- Data persistence requirements

9. Appendix
- Complete raw prompt log
- Unedited user messages only

Formatting Requirements:
- Professional PDF layout
- Clear section headings
- Page numbers
- Consistent typography
- One prompt per entry
- Preserve original intent and wording
- No assistant-generated content
- No summaries replacing prompts
- No omissions

Final Output:
Generate a clean, downloadable PDF containing every prompt and request I submitted in this conversation history, organized and structured, while preserving the original wording and chronological flow.